Abdelrahman Khalifa

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Summary

Embedded software engineer writing C firmware on an RTOS for real machines: owns the Zephyr RTOS firmware of EKSON's pneumatic motion simulators, with safety stops and a watchdog. Automotive background from an AUTOSAR-based V2X graduation project (bootloader with image verification and rollback), AUTOSAR drivers and a FreeRTOS controller with fault diagnostics.

Experience

Embedded Software Engineer Aug 2025 – Present
EKSON *Egypt*

- Wrote the ESP32-S3 firmware in C and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS for more deterministic timing and devicetree-based drivers.
- Implemented closed-loop PID control of the pneumatic valves with feedback from an MPU6050 IMU over I2C, plus safety functions: emergency stop, motion limits and a watchdog.
- Developed the desktop control app (UART protocol to the device) and cut its CPU use and latency by about 40% (measured) with asyncio.

Google Summer of Code Contributor May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) *Remote*

- Investigated and fixed crashes and a leaked client in libcups' C code, and isolated the native backend in a separate service so a crash cannot take down the app; **11 PRs merged upstream** (list).

Embedded Linux Systems Intern Dec 2024 – Jan 2025
Orange Digital Center (ODC) *Egypt*

- Built Linux images with Yocto and Buildroot and a kernel module; scored 100% in the final evaluation.

OSS Projects

Zephyr RTOS Mar 2025

- Implemented the `IP_MULTICAST_LOOP` socket option for IPv4 across 6 networking modules, merged (issue #86977).

Projects

AUTOSAR-Based V2X Vehicle | *graduation project, team; STM32, ESP32* GitHub

- Owned the firmware update path. The image is sent over UART with UDS services (request download, transfer data, transfer exit); the STM32 bootloader verifies it with the hardware CRC unit before jumping to it and keeps two banks with rollback.

Seat Heater Control System | *Tiva C, FreeRTOS* GitHub

- Two-seat real-time controller; event groups and periodic tasks, a mutex-protected shared UART display, sensor-failure detection logged to memory with a fault LED.

AUTOSAR PORT and GPIO drivers | *C, TM4C*

- MCAL drivers developed and tested against the AUTOSAR specification.

Technical Skills

Languages: C, Modern C++, Python, Rust, Bash, Verilog
Embedded & RTOS: Zephyr, FreeRTOS, OSEK/VDX (Trampoline), AUTOSAR (MCAL), bootloaders, UDS; ARM Cortex-M (STM32, TM4C), ESP32-S3, AVR
Interfaces: UART, I2C, SPI, CAN; TCP/IP sockets
Testing & tools: unit and integration testing, CI (GitHub Actions), fault diagnostics, Git, CMake, Make, debugging, Linux

Education

Minia University Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering *2020 – 2025*

- Grade: Very Good with Honors (84.6%). Codeforces Expert; 4-time ECPC finalist (2021–2024).